Question	Answer	Marks	Partial Marks
7(b)(ii)	117 or 116.9 to 117.2	**5**	**M4** for $\pi \times 2.99^2 + \pi \times 2.99 \times \sqrt{2.99^2 + (3 \times 2.99)^2}$ oe or **M3** for $\pi \times 2.99 \times \sqrt{2.99^2 + (3 \times 2.99)^2}$ or **M2** for $\sqrt{2.99^2 + (3 \times 2.99)^2}$ or **M1** for $2.99^2 + (3 \times 2.99)^2$ or for $\pi \times 2.99^2$
8(a)	Correct sketch with roots indicated at $x = -9$ and $x = 2$ and y intercept = –18 Minimum should be in 3rd quadrant	**4**	**B1** for U shaped parabola **B2** for roots at –9 and 2 on diagram or **M1** for $(x + 9)(x - 2)$ [= 0] **B1** for y – intercept at –18 on diagram Maximum 3 marks if sketch not fully correct
8(b)(i)	$2x - 3$	**2**	**B1** for $2x + k$ or $kx^{[p]} - 3$
8(b)(ii)	(1.5, –30.25) oe	**3**	**B2** for $x = 1.5$ or **M1** for *their* **(b)(i)** = 0 or for $(x - 1.5)^2$
8(c)	$x^2 - x - 33$ [= 0] seen	**B1**	
	$\frac{[--]1 \pm \sqrt{([-]1)^2 - 4(1)(-33)}}{2 \times 1}$ oe	**B2FT**	**FT** *their* quadratic dep on no factors **B1** for $\sqrt{([-]1)^2 - 4(1)(-33)}$ or better or **B1** for $\frac{[--]1 + \sqrt{q}}{2(1)}$ oe or $\frac{[--]1 - \sqrt{q}}{2(1)}$ oe
	–5.27 or –5.267 to –5.266 **and** 6.27 or 6.266 to 6.267	**B2**	**B1** for each If 0 scored, **SC1** for –6.27 and 5.27
	(–5.27, 15.53 or 15.54) **and** (6.27, –7.53 or – 7.54)	**B1**	
9(a)(i)	13	**1**	
9(a)(ii)	–20	**1**	**FT** 6 – 2(*their* **(a)(i)**)

Question	Answer	Marks	Partial Marks
9(b)	$\frac{6-x}{2}$ oe final answer	**2**	**M1** for correct first step $x = 6 - 2y$, $y - 6 = -2x$, $\frac{y}{2} = 3 - x$
9(c)	2.375 oe	**4**	**B1** for $6 - 2(2x - 7)$ oe **B1** for $4x + 1 = 6 - 4x + 14$ **M1** for $8x = 19$ FT *their* linear equation rearranged correctly from $ax + b = cx + d$ to form $ex = f$
9(d)	$\frac{1}{3}$ or 0.333…	**2**	**M1** for h(1) or 3^(3^{x-2} – 2) or 3^(3^{2-2} – 2) or better
9(e)	6561	**2**	**M1** for 3^{10-2} or x = h(10)